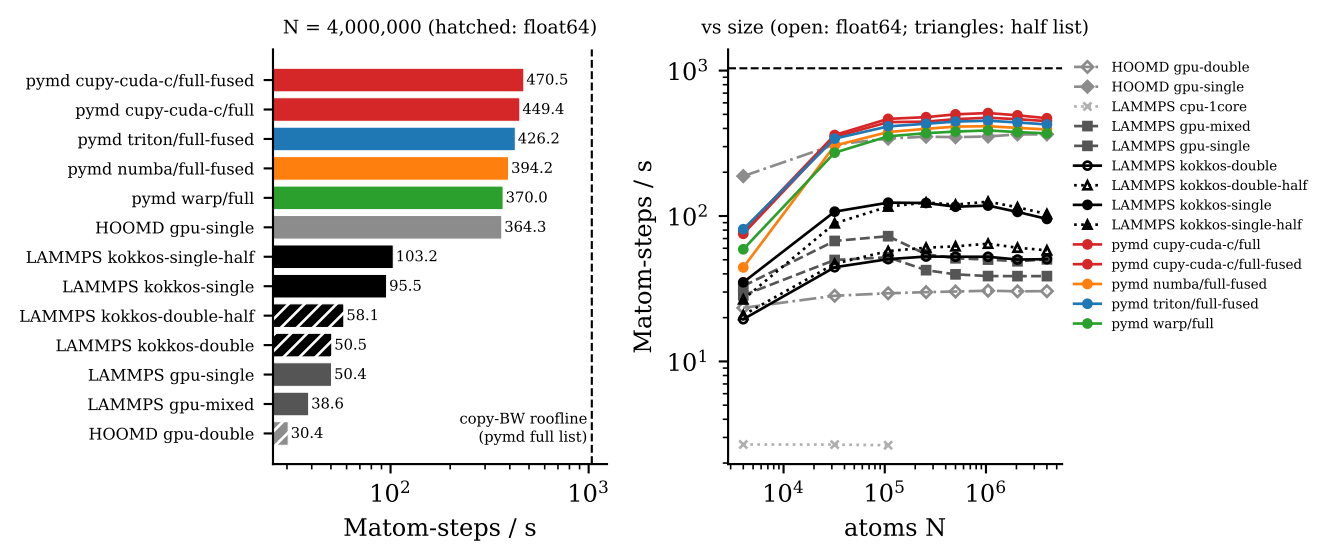
N = 4,000,000 (hatched: float64)
pymd cupy-cuda-c/full-fused
pymd cupy-cuda-c/full
pymd triton/full-fused
pymd numba/full-fused
pymd warp/full
HOOMD gpu-single
LAMMPS kokkos-single-half
LAMMPS kokkos-single
LAMMPS kokkos-double-half
LAMMPS kokkos-double
LAMMPS gpu-single
LAMMPS gpu-mixed
HOOMD gpu-double
470.5
449.4
426.2
394.2
370.0
364.3
103.2
95.5
58.1
50.5
50.4
38.6
30.4
copy-BW roofline
(pymd full list)
10^2
10^3
Matom-steps / s
vs size (open: float64; triangles: half list)
10^3
10^2
10^1
Matom-steps / s
10^4
10^5
10^6
atoms N
HOOMD gpu-double
HOOMD gpu-single
LAMMPS cpu-1core
LAMMPS gpu-mixed
LAMMPS gpu-single
LAMMPS kokkos-double
LAMMPS kokkos-double-half
LAMMPS kokkos-single
LAMMPS kokkos-single-half
pymd cupy-cuda-c/full
pymd cupy-cuda-c/full-fused
pymd numba/full-fused
pymd triton/full-fused
pymd warp/full